User Needs + Defining Success

Chapter worksheet

1. Evidence of user need

Gather existing research and make a case for using AI to solve your user need.

Social media platforms generate massive amounts of data every second. Users, businesses, and policymakers struggle to extract meaningful insights from this data due to its high volume, noise, and unstructured nature. Traditional keyword-based search tools often fall short in providing contextually relevant and insightful results. The key user needs in this domain include:

- Efficient data retrieval
- Semantic understanding
- Actionable insights
- Scalability and automation

Given these needs, AI offers a promising solution by integrating Natural Language Processing (NLP), Information Retrieval (IR), and Machine Learning (ML) techniques.

1.1.User research summary

Date	Source	Summary of Findings
2017	Vaswani et al., 2017 - Attention Is All You Need	Research in **semantic search models** (e.g., BERT, Sentence Transformers) shows significant improvements in ranking relevant content by understanding query intent
2020	Lewis et al., 2020 - Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	Studies show that **RAG significantly improves accuracy and relevance in NLP tasks** by conditioning responses on factual retrieved data
2022	Microsoft Research, 2022 - Scaling AI Workloads on Cloud Infrastructures	Cloud-based AI solutions, such as **Azure Functions for event-driven processing and Elasticsearch for indexing**, optimize large-scale data ingestion and retrieval
2014	Hutto & Gilbert, 2014 - VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of social media Text	**Sentiment analysis models** such as **RoBERTa, DistilBERT, and LSTM-based classifiers** can analyze user sentiment and trends effectively

1.2. Make a case for and against your AI feature.

Challenge	Traditional Approach	AI-Driven Approach
High data volume	Manual filtering, keyword search (slow)	Automated retrieval using Elasticsearch, Vector DB
Relevance and context understanding	Exact keyword matches	Semantic search via transformer models
Insight generation	Manual analysis	AI-generated insights via RAG models
Scalability	Hard to process millions of tweets	Cloud-based AI pipelines (Azure Functions)
Real-time analysis	Batch processing	Event-driven AI processing

Criteria	AI Probably Better	AI Probably Not Better
Efficiency & Speed	AI can process and analyze large datasets much faster than humans.	AI may generate irrelevant or misleading results without human oversight.
Scalability	AI can handle millions of tweets or data points, making it ideal for large-scale applications.	High computational cost and infrastructure needs can make AI expensive to scale.
Consistency	AI applies the same logic every time, reducing human bias and errors.	AI models can amplify biases if trained on biased datasets.
Semantic Understanding	AI-powered NLP models (e.g., BERT, GPT) provide deeper insights beyond simple keyword matching.	AI may struggle with sarcasm, humor, and nuanced discussions, leading to misinterpretation.
Automation of Tasks	AI can automate retrieval, ranking, and summarization, saving human effort.	Users may become over-reliant on AI, leading to blind trust in generated insights.
Trust & Explainability	AI-driven models can provide transparent ranking scores to justify results.	Many AI models operate as black boxes, making it hard for users to understand AI decisions.
Contextual Awareness	AI can detect patterns and trends across vast datasets.	AI lacks human intuition and may fail in highly context-dependent scenarios.
Cost & Maintenance	AI can provide long-term cost savings by reducing manual effort.	AI models require ongoing updates, monitoring, and high computing resources.

1.3. Can AI solve this problem in a unique way?

Research-backed evidence shows that AI-driven techniques such as Retrieval-Augmented Generation (RAG), Elasticsearch, and Transformer-based models significantly enhance data retrieval, contextual search, and insight generation from unstructured social media text.

By integrating AI in MLOps for social media analytics, we ensure:

1. High-performance data retrieval
2. Accurate and context-aware insights
3. Scalability to handle millions of tweets.

4. Real-time query responsiveness

Thus, AI is the most efficient, scalable, and reliable solution for extracting insights from large-scale social media data.

2. Augmentation vs. Automation

We hypothesize that AI is a good fit for social media insights retrieval and analysis, but users prefer an augmented approach rather than full automation.

- AI should handle repetitive, data-heavy tasks (retrieval, ranking, filtering).
- Users should have control over final insights (validation, editing, context refinement).

Task	Preferred AI Role	Automation vs. Augmentation
Retrieving tweets for analysis	AI retrieves and ranks tweets.	Automation (fully AI-driven)
Filtering & categorizing insights	AI suggests, but users refine categories.	Augmentation (human-in-the-loop)
Generating summaries (RAG Model)	AI generates drafts, users edit.	Augmentation (co-editing model)
Trend detection & sentiment analysis	AI surfaces patterns, users validate trends.	Augmentation (AI assists, but users confirm trends)
Final reporting & decision-making	AI provides insights, users refine and present findings.	Augmentation (human control over final output)

User Need	Best AI Approach	Why?
Speed & efficiency	Automation	AI should handle data-heavy retrieval and filtering.
Accuracy & trust	Augmentation	Users prefer control over final insights.
Scalability	Automation	AI can process vast amounts of social media data.
Context awareness	Augmentation	AI may miss nuances, sarcasm, and cultural factors.

3. Design your reward function.

3.1.Optimization Strategy: Precision vs. Recall

After analyzing user attitudes towards **automation vs. augmentation**, we have identified key trade-offs in our AI-powered **social media insights system**.

3.2.Reward Function Template

Our AI model will be optimized for precision.

because users prefer highly relevant insights with minimal noise. Our research shows that users are more frustrated by false positives (irrelevant or misleading results) than by missing a few relevant ones. Ensuring high precision allows users to trust AI-generated insights without requiring extensive manual filtering.

Trade-offs and User Impact

We understand that the trade-off for choosing this method means our model will occasionally miss some relevant insights (lower recall).

- Users may need to refine search queries or re-run searches to discover missed insights.
- Exploratory analysis users (e.g., researchers, trend analysts) may feel limited in retrieving all possible relevant tweets.
- However, users will spend less time manually filtering irrelevant or misleading tweets, improving efficiency.

3.3.Implementation Strategy

- Threshold tuning: Set a high confidence threshold for relevance scoring to prioritize precision.
- User-adjustable recall settings: Offer an optional "broad mode" for users who prefer higher recall.
- Feedback-driven ranking: Allow users to flag irrelevant results to continuously improve precision.

By optimizing for precision, we enhance trust, relevance, and user satisfaction, while still allowing flexibility for users who need broader recall-based exploration.

4. Defining Success Criteria

Now that we have determined that AI is a good fit for our social media insights retrieval system and have chosen to optimize for precision, we will define clear success criteria to measure our AI's effectiveness.
These success metrics will help us monitor performance, identify failures, and iterate as needed.

4.1. Success Metrics Framework

Version	Success Metric Statement
Version 1	If precision drops below 80% for our AI-driven social media insights retrieval system, we will review the ranking model, retrain with better datasets, and adjust relevance thresholds.
Version 2	If query latency goes above 2 seconds on average for user searches, we will optimize Elasticsearch queries, scale infrastructure, and introduce caching mechanisms.
Version 3	If user satisfaction (measured through feedback ratings) falls below 70%, we will analyze flagged results, improve explainability, and introduce user-adjustable ranking options.

Metric Type	**Success Metric Statement**	**Threshold**	**Action to Take if Threshold is Not Met**
Precision-Based Success	If precision drops below 80% for our AI-driven insights system	< 80%	Review ranking model, retrain AI with better datasets, adjust relevance thresholds
Latency-Based Success	If query latency exceeds 2 seconds on average for user searches	> 2 sec	Optimize Elasticsearch queries, introduce caching, scale infrastructure
User Satisfaction-Based Success	If user satisfaction (measured via feedback ratings) falls below 70%	< 70%	Analyze flagged results, improve explainability, introduce user-adjustable ranking options
Recall-Based Success	If recall drops below 60%, meaning too many relevant tweets are missed	< 60%	Fine-tune the AI's ranking model, adjust query parameters, add an optional "broad search" mode
System Uptime	If system uptime drops below 99% over a 30-day period	< 99% uptime	Scale cloud infrastructure, improve failure recovery processes
User Engagement	If click-through rate (CTR) on AI-generated insights falls below 50%	< 50% CTR	Optimize insight summaries, enhance UI/UX for better readability

4.2. Success Metric Review

Review Schedule	Attendees
First Month: Data Lake & Blob Setup, Data Preprocessing Dev	Whole team
Second Month: Elasticsearch Integration, RAG Model (Azure Function), Frontend & Backend Services	Whole team
Third Month: CI/CD Pipeline Completion	Whole team
Forth Month: Final Testing, Documentation, & Presentation	Whole team, Professor, and Guests